

SARAH BASHANDY

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EDUCATION

University of California, Los Angeles

Sep 2023 - Jun 2026

B.S, Electrical Engineering

• **GPA:** 3.59

University of California, Los Angeles

Jun 2027

MS, Electrical and Computer Engineering (Part time)

WORK EXPERIENCE

Wing (Google X)

Jun 2026 - Sep 2026

Hardware reliability intern

Palo Alto, CA

- Developed an accelerometer instrumentation method for drones to capture vibration data at critical test points during flight testing.
- Planned and oversaw flight tests, creating test matrices, tracking test conditions, validating datasets, and identifying anomalies and exceptions.
- Processed flight vibration data and developed accelerated lifetime test profiles using MIL-STD-810H, identifying frequency-dependent vibration levels across flight phases and test points.
- Post-processed vibration data for the Aero team to support validation of flexible-body simulation models.

UCLA

Jun 2025 - Aug 2025

UCLA Academic group Tutor, ENG 1: Autonomous rovers

- Delivered instruction to classes of 20 students on digital systems concepts, circuit wiring, embedded programming (Arduino/C++), and schematic capture, integrating electrical circuit design analysis principles.
- Mentored students in design for manufacturability and iterative testing of autonomous rover systems, emphasizing electrical design and circuit analysis.

PROJECT AND DESIGN EXPERIENCE

Design and Tapeout of 8 bit synchronous SAR ADC using TSMC 180nm technology

Jan 2026 - Present

- Designed and simulated 3 blocks using schematic view and Verilog-A - Strong arm latch comparator, SAR Logic block, 8-bit CDAC.
- Implemented transistor-level schematic and layout for key blocks in Cadence Virtuoso, targeting a compact layout area under $400\text{ }\mu\text{m} \times 400\text{ }\mu\text{m}$.
- Characterized dynamic performance by calculating SNDR, SFDR, ENOB (7.3 bits), and power consumption (core design: 6-7uW) ensuring the clock distribution system met jitter requirements for high-speed sampling.
- Developed a verification PCB for chip-level testing, implementing layout techniques to reduce parasitic inductance and improve signal integrity for high-speed board bring-up

Electrical Team CubeSat project

Sep 2024 - Present

Bruin Spacecraft Group

- Performing voltage drop analysis and PCB verification on Altium to ensure signal integrity and power reliability for EPS PCB.
- Supporting the design and testing of antenna systems for a 3U cube satellite missions.
- Routing and layout for four layer and two layer PCBs

Micromouse

Aug 2024 - Jun 2025

IEEE UCLA

- Developed a PCB-based robot for maze-solving using embedded control systems (STM32).
- Gained hands-on experience with PCB design, soldering and embedded programming using microcontrollers.
- Programmed and optimized algorithms for autonomous navigation.

Smartwatch LCD Display driver (IC design)

Aug 2025 - Dec 2025

- Designed a two-stage amplifier, featuring a Telescopic Cascode input stage and a Class AB Common Source output stage on Cadence using TSMC 45nm technology (gpdk045).
- Achieved approximately 3000 open loop gain with the first stage providing 370x gain. Managed design constraints to ensure stability (Phase Margin greater than 45%) and met the strict power consumption limit (under 1.25 mW) while optimizing for settling time.

SKILLS

- **Software & Tools:** Altium, Kicad, LTspice, SolidWorks, Cube IDE, Cadence, GitHub, Ham radio license, SDR, Electrical Simulation
- **PCB design and simulation:** SPICE (LTspice), Ansys HFSS, Spectre, Verilog-A, Cadence Virtuoso
- **Programming:** C++, Python, MATLAB, embedded systems (Arduino, ESP32, STM32, Teensy)
- **Testing And Verification:** Oscilloscope, Vector network analyzer, spectrum analyzer, DMM, circuit debugging, Simulink, Monte Carlo (Mismatch/Process), Corner Analysis (PVT), Post-Layout Extraction (PEX), SNDR/SFDR Characterization